Global Event Query / LoRA / Hard-RoI ablation: five backbones, four configurations (tau = 0.4 / 0.5 / 0.6)

Test split: 1,036 queries (544 positive / 492 negative); all values: %. Parentheses: gain versus the same-backbone GMR, pp.

		Operating threshold tau = 0.4								
		Null-Set Rejection			*Temporal Localization*				*Overall GMR*	
Backbone	**Configuration (all include GMR)**	N-acc	AUROC	Rej-F1	mAP	mR@1	mR@5	mR+@5	G-mIoU@1	G-mIoU@3
Moment-DETR	1. + Global Event Query	**100.00** (+100.00)	**70.33** (+0.13)	**75.00** (+75.00)	6.68 (+0.00)	**4.64** (+0.00)	9.44 (+0.00)	0.88 (+0.00)	52.02 (+46.19)	49.00 (+46.57)
	2. + Global Event Query + LoRA	**100.00** (+100.00)	70.32 (+0.12)	**75.00** (+75.00)	5.87 (-0.81)	4.01 (-0.63)	8.17 (-1.27)	0.91 (+0.03)	51.78 (+45.95)	48.92 (+46.49)
	3. + Global Event Query + Hard-RoI	**100.00** (+100.00)	70.22 (+0.02)	**75.00** (+75.00)	5.38 (-1.30)	3.18 (-1.46)	7.54 (-1.90)	**1.30** (+0.42)	52.22 (+46.39)	49.07 (+46.64)
	4. + Global Event Query + LoRA + Hard-RoI	**100.00** (+100.00)	70.10 (-0.10)	**75.00** (+75.00)	**7.35** (+0.67)	3.48 (-1.16)	**11.13** (+1.69)	0.76 (-0.12)	**52.34** (+46.51)	**49.11** (+46.68)
QD-DETR	1. + Global Event Query	80.28 (+23.57)	76.44 (-0.57)	71.11 (+9.38)	14.62 (+0.00)	9.47 (+0.00)	20.17 (+0.00)	7.43 (+0.00)	46.60 (+10.09)	41.78 (+10.31)
	2. + Global Event Query + LoRA	80.49 (+23.78)	**76.48** (-0.53)	71.22 (+9.49)	13.71 (-0.91)	8.56 (-0.91)	19.67 (-0.50)	6.54 (-0.89)	46.00 (+9.49)	41.58 (+10.11)
	3. + Global Event Query + Hard-RoI	81.91 (+25.20)	76.26 (-0.75)	71.84 (+10.11)	14.99 (+0.37)	9.18 (-0.29)	21.73 (+1.56)	7.60 (+0.17)	47.32 (+10.81)	42.52 (+11.05)
	4. + Global Event Query + LoRA + Hard-RoI	**84.35** (+27.64)	76.21 (-0.80)	**72.43** (+10.70)	**16.49** (+1.87)	**10.04** (+0.57)	**23.92** (+3.75)	**11.11** (+3.68)	**48.38** (+11.87)	**43.64** (+12.17)
CG-DETR	1. + Global Event Query	**89.23** (+66.06)	76.70 (+0.30)	**73.78** (+38.32)	18.66 (+0.00)	12.65 (+0.00)	24.90 (+0.00)	11.98 (+0.00)	**50.55** (+27.71)	**45.67** (+28.60)
	2. + Global Event Query + LoRA	87.80 (+64.63)	**76.73** (+0.33)	72.97 (+37.51)	18.34 (-0.32)	12.17 (-0.48)	24.72 (-0.18)	11.88 (-0.10)	49.69 (+26.85)	44.96 (+27.89)
	3. + Global Event Query + Hard-RoI	86.18 (+63.01)	**76.73** (+0.33)	72.60 (+37.14)	**19.05** (+0.39)	12.66 (+0.01)	**25.49** (+0.59)	12.88 (+0.90)	49.25 (+26.41)	44.38 (+27.31)
	4. + Global Event Query + LoRA + Hard-RoI	87.80 (+64.63)	76.64 (+0.24)	73.03 (+37.57)	18.93 (+0.27)	**13.19** (+0.54)	25.09 (+0.19)	**13.10** (+1.12)	50.02 (+27.18)	45.00 (+27.93)
EaTR	1. + Global Event Query	82.52 (+33.94)	75.73 (-0.83)	71.35 (+14.24)	16.22 (+0.00)	10.64 (+0.00)	21.42 (+0.00)	9.59 (+0.00)	47.14 (+13.82)	42.52 (+14.18)
	2. + Global Event Query + LoRA	83.94 (+35.36)	75.76 (-0.80)	71.76 (+14.65)	16.45 (+0.23)	10.96 (+0.32)	21.55 (+0.13)	9.37 (-0.22)	47.78 (+14.46)	43.18 (+14.84)
	3. + Global Event Query + Hard-RoI	82.32 (+33.74)	75.91 (-0.65)	71.55 (+14.44)	16.06 (-0.16)	10.47 (-0.17)	21.46 (+0.04)	**10.92** (+1.33)	47.09 (+13.77)	42.48 (+14.14)
	4. + Global Event Query + LoRA + Hard-RoI	**85.98** (+37.40)	**76.02** (-0.54)	**72.12** (+15.01)	**17.16** (+0.94)	**11.43** (+0.79)	**23.19** (+1.77)	9.55 (-0.04)	**48.71** (+15.39)	**44.08** (+15.74)
FlashVTG	1. + Global Event Query	88.01 (+55.29)	75.26 (+0.96)	73.08 (+28.67)	25.40 (-0.03)	15.11 (-0.06)	34.94 (+0.00)	17.86 (+0.00)	50.48 (+22.38)	45.43 (+23.68)
	2. + Global Event Query + LoRA	88.82 (+56.10)	**75.32** (+1.02)	73.45 (+29.04)	25.42 (-0.01)	15.27 (+0.10)	35.45 (+0.51)	17.54 (-0.32)	50.80 (+22.70)	45.74 (+23.99)
	3. + Global Event Query + Hard-RoI	**90.65** (+57.93)	75.29 (+0.99)	**73.72** (+29.31)	24.20 (-1.23)	14.01 (-1.16)	33.62 (-1.32)	17.08 (-0.78)	51.60 (+23.50)	**46.57** (+24.82)
	4. + Global Event Query + LoRA + Hard-RoI	90.04 (+57.32)	75.25 (+0.95)	73.34 (+28.93)	**26.24** (+0.81)	**16.23** (+1.06)	**36.27** (+1.33)	**18.27** (+0.41)	**51.67** (+23.57)	46.41 (+24.66)

		Operating threshold tau = 0.5								
		Null-Set Rejection			*Temporal Localization*				*Overall GMR*	
Backbone	**Configuration (all include GMR)**	N-acc	AUROC	Rej-F1	mAP	mR@1	mR@5	mR+@5	G-mIoU@1	G-mIoU@3
Moment-DETR	1. + Global Event Query	**100.00** (+20.73)	**70.33** (+0.13)	**75.00** (+7.06)	6.68 (+0.00)	**4.64** (+0.00)	9.44 (+0.00)	0.88 (+0.00)	52.02 (+9.52)	49.00 (+9.61)
	2. + Global Event Query + LoRA	**100.00** (+20.73)	70.32 (+0.12)	**75.00** (+7.06)	5.87 (-0.81)	4.01 (-0.63)	8.17 (-1.27)	0.91 (+0.03)	51.78 (+9.28)	48.92 (+9.53)
	3. + Global Event Query + Hard-RoI	**100.00** (+20.73)	70.22 (+0.02)	**75.00** (+7.06)	5.38 (-1.30)	3.18 (-1.46)	7.54 (-1.90)	**1.30** (+0.42)	52.22 (+9.72)	49.07 (+9.68)
	4. + Global Event Query + LoRA + Hard-RoI	**100.00** (+20.73)	70.10 (-0.10)	**75.00** (+7.06)	**7.35** (+0.67)	3.48 (-1.16)	**11.13** (+1.69)	0.76 (-0.12)	**52.34** (+9.84)	**49.11** (+9.72)
QD-DETR	1. + Global Event Query	92.48 (+13.01)	76.44 (-0.57)	74.29 (+2.81)	14.62 (+0.00)	9.47 (+0.00)	20.17 (+0.00)	7.43 (+0.00)	51.54 (+5.05)	46.91 (+5.44)
	2. + Global Event Query + LoRA	92.48 (+13.01)	**76.48** (-0.53)	74.41 (+2.93)	13.71 (-0.91)	8.56 (-0.91)	19.67 (-0.50)	6.54 (-0.89)	51.00 (+4.51)	46.73 (+5.26)
	3. + Global Event Query + Hard-RoI	92.68 (+13.21)	76.26 (-0.75)	74.45 (+2.97)	14.99 (+0.37)	9.18 (-0.29)	21.73 (+1.56)	7.60 (+0.17)	51.67 (+5.18)	46.96 (+5.49)
	4. + Global Event Query + LoRA + Hard-RoI	**93.90** (+14.43)	76.21 (-0.80)	**74.58** (+3.10)	**16.49** (+1.87)	**10.04** (+0.57)	**23.92** (+3.75)	**11.11** (+3.68)	**52.16** (+5.67)	**47.48** (+6.01)
CG-DETR	1. + Global Event Query	**96.95** (+46.54)	76.70 (+0.30)	75.36 (+17.89)	18.66 (+0.00)	12.65 (+0.00)	24.90 (+0.00)	11.98 (+0.00)	53.49 (+19.14)	48.76 (+19.62)
	2. + Global Event Query + LoRA	**96.95** (+46.54)	**76.73** (+0.33)	75.42 (+17.95)	18.34 (-0.32)	12.17 (-0.48)	24.72 (-0.18)	11.88 (-0.10)	53.41 (+19.06)	48.74 (+19.60)
	3. + Global Event Query + Hard-RoI	**96.95** (+46.54)	**76.73** (+0.33)	**75.53** (+18.06)	**19.05** (+0.39)	12.66 (+0.01)	**25.49** (+0.59)	12.88 (+0.90)	53.65 (+19.30)	**48.86** (+19.72)
	4. + Global Event Query + LoRA + Hard-RoI	**96.95** (+46.54)	76.64 (+0.24)	75.36 (+17.89)	18.93 (+0.27)	**13.19** (+0.54)	25.09 (+0.19)	**13.10** (+1.12)	**53.67** (+19.32)	48.80 (+19.66)
EaTR	1. + Global Event Query	95.53 (+28.86)	75.73 (-0.83)	**75.26** (+9.66)	16.22 (+0.00)	10.64 (+0.00)	21.42 (+0.00)	9.59 (+0.00)	52.83 (+11.98)	48.29 (+12.29)
	2. + Global Event Query + LoRA	95.53 (+28.86)	75.76 (-0.80)	74.96 (+9.36)	16.45 (+0.23)	10.96 (+0.32)	21.55 (+0.13)	9.37 (-0.22)	52.81 (+11.96)	48.22 (+12.22)
	3. + Global Event Query + Hard-RoI	95.53 (+28.86)	75.91 (-0.65)	75.20 (+9.60)	16.06 (-0.16)	10.47 (-0.17)	21.46 (+0.04)	**10.92** (+1.33)	52.69 (+11.84)	48.23 (+12.23)
	4. + Global Event Query + LoRA + Hard-RoI	**95.93** (+29.26)	**76.02** (-0.54)	74.98 (+9.38)	**17.16** (+0.94)	**11.43** (+0.79)	**23.19** (+1.77)	9.55 (-0.04)	**52.95** (+12.10)	**48.34** (+12.34)
FlashVTG	1. + Global Event Query	94.72 (+23.79)	75.26 (+0.96)	74.38 (+8.03)	25.40 (-0.03)	15.11 (-0.06)	34.94 (+0.00)	17.86 (+0.00)	53.01 (+9.44)	48.11 (+9.84)
	2. + Global Event Query + LoRA	94.31 (+23.38)	**75.32** (+1.02)	74.18 (+7.83)	25.42 (-0.01)	15.27 (+0.10)	35.45 (+0.51)	17.54 (-0.32)	52.78 (+9.21)	47.90 (+9.63)
	3. + Global Event Query + Hard-RoI	**95.93** (+25.00)	75.29 (+0.99)	**74.80** (+8.45)	24.20 (-1.23)	14.01 (-1.16)	33.62 (-1.32)	17.08 (-0.78)	53.51 (+9.94)	48.61 (+10.34)
	4. + Global Event Query + LoRA + Hard-RoI	95.73 (+24.80)	75.25 (+0.95)	74.64 (+8.29)	**26.24** (+0.81)	**16.23** (+1.06)	**36.27** (+1.33)	**18.27** (+0.41)	**53.68** (+10.11)	**48.62** (+10.35)

		Operating threshold tau = 0.6								
		Null-Set Rejection			*Temporal Localization*				*Overall GMR*	
Backbone	**Configuration (all include GMR)**	N-acc	AUROC	Rej-F1	mAP	mR@1	mR@5	mR+@5	G-mIoU@1	G-mIoU@3
Moment-DETR	1. + Global Event Query	**100.00** (+0.00)	**70.33** (+0.13)	**75.00** (+0.06)	6.68 (+0.00)	**4.64** (+0.00)	9.44 (+0.00)	0.88 (+0.00)	52.02 (+0.00)	49.00 (+0.00)
	2. + Global Event Query + LoRA	**100.00** (+0.00)	70.32 (+0.12)	**75.00** (+0.06)	5.87 (-0.81)	4.01 (-0.63)	8.17 (-1.27)	0.91 (+0.03)	51.78 (-0.24)	48.92 (-0.08)
	3. + Global Event Query + Hard-RoI	**100.00** (+0.00)	70.22 (+0.02)	**75.00** (+0.06)	5.38 (-1.30)	3.18 (-1.46)	7.54 (-1.90)	**1.30** (+0.42)	52.22 (+0.20)	49.07 (+0.07)
	4. + Global Event Query + LoRA + Hard-RoI	**100.00** (+0.00)	70.10 (-0.10)	**75.00** (+0.06)	**7.35** (+0.67)	3.48 (-1.16)	**11.13** (+1.69)	0.76 (-0.12)	**52.34** (+0.32)	**49.11** (+0.11)
QD-DETR	1. + Global Event Query	96.54 (+5.89)	76.44 (-0.57)	75.04 (+1.75)	14.62 (+0.00)	9.47 (+0.00)	20.17 (+0.00)	7.43 (+0.00)	53.09 (+2.41)	48.44 (+2.40)
	2. + Global Event Query + LoRA	96.54 (+5.89)	**76.48** (-0.53)	75.16 (+1.87)	13.71 (-0.91)	8.56 (-0.91)	19.67 (-0.50)	6.54 (-0.89)	52.54 (+1.86)	48.26 (+2.22)
	3. + Global Event Query + Hard-RoI	96.75 (+6.10)	76.26 (-0.75)	75.14 (+1.85)	14.99 (+0.37)	9.18 (-0.29)	21.73 (+1.56)	7.60 (+0.17)	53.20 (+2.52)	48.54 (+2.50)
	4. + Global Event Query + LoRA + Hard-RoI	**97.56** (+6.91)	76.21 (-0.80)	**75.47** (+2.18)	**16.49** (+1.87)	**10.04** (+0.57)	**23.92** (+3.75)	**11.11** (+3.68)	**53.70** (+3.02)	**48.96** (+2.92)
CG-DETR	1. + Global Event Query	98.98 (+20.52)	76.70 (+0.30)	75.68 (+4.59)	18.66 (+0.00)	12.65 (+0.00)	24.90 (+0.00)	11.98 (+0.00)	54.34 (+7.68)	49.62 (+8.09)
	2. + Global Event Query + LoRA	98.98 (+20.52)	**76.73** (+0.33)	75.68 (+4.59)	18.34 (-0.32)	12.17 (-0.48)	24.72 (-0.18)	11.88 (-0.10)	54.25 (+7.59)	49.59 (+8.06)
	3. + Global Event Query + Hard-RoI	**99.19** (+20.73)	**76.73** (+0.33)	**75.78** (+4.69)	**19.05** (+0.39)	12.66 (+0.01)	**25.49** (+0.59)	12.88 (+0.90)	**54.58** (+7.92)	**49.79** (+8.26)
	4. + Global Event Query + LoRA + Hard-RoI	**99.19** (+20.73)	76.64 (+0.24)	75.66 (+4.57)	18.93 (+0.27)	**13.19** (+0.54)	25.09 (+0.19)	**13.10** (+1.12)	54.54 (+7.88)	49.73 (+8.20)
EaTR	1. + Global Event Query	98.98 (+11.78)	75.73 (-0.83)	75.68 (+2.22)	16.22 (+0.00)	10.64 (+0.00)	21.42 (+0.00)	9.59 (+0.00)	54.22 (+4.66)	49.56 (+4.70)
	2. + Global Event Query + LoRA	98.98 (+11.78)	75.76 (-0.80)	75.56 (+2.10)	16.45 (+0.23)	10.96 (+0.32)	21.55 (+0.13)	9.37 (-0.22)	54.24 (+4.68)	49.57 (+4.71)
	3. + Global Event Query + Hard-RoI	**99.19** (+11.99)	75.91 (-0.65)	**75.72** (+2.26)	16.06 (-0.16)	10.47 (-0.17)	21.46 (+0.04)	**10.92** (+1.33)	54.16 (+4.60)	**49.65** (+4.79)
	4. + Global Event Query + LoRA + Hard-RoI	**99.19** (+11.99)	**76.02** (-0.54)	75.60 (+2.14)	**17.16** (+0.94)	**11.43** (+0.79)	**23.19** (+1.77)	9.55 (-0.04)	**54.25** (+4.69)	**49.65** (+4.79)
FlashVTG	1. + Global Event Query	96.95 (+11.79)	75.26 (+0.96)	74.59 (+3.03)	25.40 (-0.03)	15.11 (-0.06)	34.94 (+0.00)	17.86 (+0.00)	53.71 (+4.30)	48.90 (+4.60)
	2. + Global Event Query + LoRA	96.75 (+11.59)	**75.32** (+1.02)	**74.67** (+3.11)	25.42 (-0.01)	15.27 (+0.10)	35.45 (+0.51)	17.54 (-0.32)	53.68 (+4.27)	48.83 (+4.53)
	3. + Global Event Query + Hard-RoI	**97.36** (+12.20)	75.29 (+0.99)	74.61 (+3.05)	24.20 (-1.23)	14.01 (-1.16)	33.62 (-1.32)	17.08 (-0.78)	53.93 (+4.52)	49.09 (+4.79)
	4. + Global Event Query + LoRA + Hard-RoI	**97.36** (+12.20)	75.25 (+0.95)	74.61 (+3.05)	**26.24** (+0.81)	**16.23** (+1.06)	**36.27** (+1.33)	**18.27** (+0.41)	**54.19** (+4.78)	**49.20** (+4.90)

Rows: Global Event Query; Global + LoRA; Global + Hard-RoI; Global + LoRA + Hard-RoI. All start independently from the same pretrained GMR.

Every cell: value (delta versus the same-backbone GMR, same Test / threshold), pp. GMR reference is not displayed as a separate row.

Bold: within-backbone best among these four rows; underline: second distinct best (ties retained). Fixed adaptation epoch 4; seed 2026.

Gaussian-RoI, 6D saliency and candidate presence fusion: OFF. Native FlashVTG calibration / rounding / NMS retained. Single-seed exploratory results.